

英 問

英 語

令 和 6 年 度 (前期)

注 意

1. 「解答はじめ」というまで、この問題冊子を開いてはいけない。
2. 問題は1冊(本文12ページ)、解答用紙は4枚である。
3. 全部の解答用紙に受験番号を書くこと。受験番号は次の要領で明確に記入する。

(例) 受験番号 80001 番の場合 →

8	0	0	0	1
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4. 解答は解答用紙の所定の位置に書くこと。他の所に書くと無効になることがある。解答用紙の裏面は使ってはならない。字数などの指示がある場合は、その指示に従って書くこと。解答文は横書きとする。
5. 書き損じても、代わりの用紙は交付しない。
6. 試験終了後、問題冊子は持ち帰ること。

I 次の英文を読み、下の問いに答えなさい。（*を付した語句には、問題文の末尾に注がある。）

In the British education system in recent decades, students are often forced to choose between studying either math and science or studying the humanities*. I recall that at the end of my very last English class at high school, in 1991, the teacher gave me a lovely handwritten note ⁽¹⁾ with a long list of books she thought I might like, saying, “Sorry to lose you to the lab*.” I was sorry that she considered me lost — because I wasn’t. I love language; I love the way words fit together; I love the way that fiction — like mathematics — can create, play with, and test the limits of imaginary worlds. I went off to Oxford University to study mathematics, very happy to be living one street away from the pub where my childhood literary heroes C. S. Lewis and J. R. R. Tolkien had met each week to discuss their work. The idea that one would have to choose between mathematics and literature is, I think, something of a tragedy — not only because the two fields are inseparably and fundamentally linked, but also because understanding these links can enhance your enjoyment of both.

We can find mathematics at the heart of literature. The universe is full of underlying structure, pattern, and regularity, and mathematics is the best tool we have for understanding it. That’s why mathematics is often called the language of the universe, and why it is so vital to science. Since we humans are part of the universe, it is only natural that our forms of creative expression, literature among them, will also manifest an inclination for pattern and structure. That ⁽²⁾ feeling we get when we read a great novel or a perfect poem — that here is a beautiful thing, with all the parts fitting together perfectly in a harmonious whole — is the same feeling a mathematician experiences when reading a beautiful proof.

All writing has structure. Letters form words, words form sentences, sentences form paragraphs. Just like the point-line-plane hierarchy of geometry*,

rules can be imposed at any stage of writing. The question is not whether to have a structure, but what structure to choose. The French author Georges Perec once wrote a novel without once using the letter “e”. Eleanor Catton’s prize-winning novel, *The Luminaries*, imposes a precise numerical rule on its chapters, each of which is half the length of the last. In both cases, ⁽³⁾the structural restrictions echo and enforce the novels’ themes. The restrictions we choose inspire us to create, to see what is possible — and it’s just the same in math.

It’s worth pointing out as well that the links between mathematics and literature do not run in just one direction. Mathematics itself has a rich tradition of linguistic creativity. Going back to early India, ⁽⁴⁾Sanskrit* mathematics followed an oral tradition*. Mathematical rules were converted to poetry so that they could be passed on by word of mouth. We think of mathematical concepts as relating to precise, fixed words: square, circle. But in the Sanskrit tradition, your words must fit into the meter* of your poem. Number words, for example, can be replaced with words for relevant objects. The number one can be represented by anything that is unique, like the Moon or the Earth, while “hand” can mean two, because we have two hands — but so can “black and white”, because that forms a pair. The expression “three voids* teeth” doesn’t mean a visit to the dentist, but that three zeros should follow the number of teeth we have: a poetic way to say 32,000. The huge variety of words and meanings lends a wonderful richness to the mathematics.

Just as mathematics makes use of literary metaphors, literature abounds with ideas that a mathematically trained eye can detect and explore. This adds an extra dimension to our appreciation of a work of fiction. As the pioneering Russian mathematician Sofia Kovalevskaya wrote, “It is impossible to be a mathematician without being a poet in soul . . . the poet must see what others do not see, must see more deeply . . . and the mathematician must do the same.” By seeing mathematics and literature as part of the same quest — to understand the

world and our place in it — we can add to our experience of both, and bring whole new kinds of enjoyment to our favorite writing.

注 humanities 人文学

lab (主に理数系の)実験室・研究所(laboratory の略)

geometry 幾何学

Sanskrit (古代インドの)サンスクリット語・文化の

oral tradition 口頭による伝承

meter 韻律

void 空白

- 1 下線部(1)について、①高校の先生から受け取った手書きのメッセージの意味と、②それについて筆者が思ったことを、それぞれ 50 字以内の日本語(句読点を含む)で説明しなさい。
- 2 下線部(2)を和訳しなさい。
- 3 下線部(3)が指している具体的な 2 つの事柄を、それぞれ 20 字以内の日本語(句読点を含む)で説明しなさい。ただし、固有名詞は使用しないこと。
- 4 下線部(4)において、数字「1」と「2」はどのようなもので表されるか、それぞれ 30 字以内の日本語(句読点を含む)で説明しなさい。
- 5 文学の中に数学を見いだすことができる理由について、筆者の考えを 100 字以内の日本語(句読点を含む)で説明しなさい。

Ⅱ 次の英文を読み、下の問いに答えなさい。（*を付した語句には、問題文の末尾に注がある。）

It's finally the weekend. This is your opportunity to rest and recharge your batteries from an exhausting work week. There is one problem, though. Relaxing is hard work in itself. You feel restless every time you sit down to take a break. But there's an irresistible urge to turn on the laptop and prepare for work on Monday.

What is going on? How can you look forward to the weekend (A) to struggle to detach from work? To (B) understand this paradox, we need to explore a fundamental concept in psychology: Unsatisfied needs influence your thoughts and actions.

This is apparent with physical needs. Imagine you are hungry, thirsty, or need to use the toilet but are stuck in traffic. Every second feels (イ) an eternity as you think about satisfying these needs. The same holds true for emotional needs. According to psychologist Abraham Maslow, needs are organized in a hierarchy with complex emotional needs emerging after basic physical needs have been satisfied. These include the need to feel safe, to belong, earn respect, and reach our fullest potential.

The psychologist David McClelland argued that additional needs are fundamental to human motivation including the need to achieve, which represents a drive to feel competent as you compete against a task, yourself, and others. Take a moment to consider how your need to achieve developed from a young age. When you were in school, you worked hard to earn good grades, make the soccer team, or perform at the school play. Parents and teachers praised you for performing well. Receiving approval and awards for a job well done only reinforced your need to achieve.

Your need to achieve is a chameleon that manifests differently based on your values. If you care about accumulating power, you will work tirelessly to

climb the corporate ladder. If you care about wealth, then your goal will be to make a huge amount of money and buy fancy stuff. If you are academically inclined, your focus is to boost your resume* with publications and presentations to stand (C) from peers. Sometimes the need to achieve is more subtle. If you prefer to stay at home, you may care most about having the best kept lawn in the neighborhood.

There is nothing wrong with setting and achieving goals as long as you are mindful of your motives and not completely consumed by your pursuits. The problem occurs when your need to achieve becomes excessive and overwhelms other parts of your life. Any time a need becomes excessive, you will develop a problem. This is apparent with physical needs. If you need to consume a hot fudge sundae* every night, it is only a matter of time before your health suffers. If you need 18 hours of sleep to have sufficient energy, then you are sleeping your life (D).

The same holds true for emotional needs. If you are desperate to be in a relationship and ignore warning signs, then you will settle for the wrong partner. If your need for safety is such that you are too afraid to leave your house, again you have a problem.

As society becomes increasingly more competitive and achievement-focused, it is easy to be swept by the chase to achieve more. Don't sacrifice your health and loved ones at the altar of achievement. Work (□) having a healthy relationship with achievement.

Here are four steps to master your need to achieve:

1. Redefine Success

Society defines success based on external standards that are easily viewed by many, such as fame and fortune. It idealizes those who excel (∧) these areas. Resist such social expectations (二) focusing on internal, more intimate, standards, such as spending quality time with loved ones, being a good person, and doing a kind deed for the day. These standards may go unnoticed by

most but have a positive impact in your life and the lives of those who matter most to you.

2. Avoid Social Comparisons

Social comparisons make you feel the urge to do more. They are also a path to suffering. Comparing your real, messy life to someone's idealized image of their life does not serve you. Focus (E) on your personal journey. Concentrate on where you have been, where you are, and where you are heading. After all, we are pursuing different goals to fulfill different emotional needs.

3. Ask Yourself: Why

Before you start a journey towards a goal, press “pause” and reflect on your motives. (あ). Make sure you have realistic expectations of its potential impact on you. We often idealize success and overestimate its impact on happiness. The reality is that achievement comes with its fair share of challenges, such as increased responsibility, stress, and scrutiny. Having a complete picture will protect you from the trap of having unrealistic expectations.

4. Rest is Part of the Process

Taking a break can be hard when you have so many tasks to achieve but so little time. The natural reflex action is to push yourself beyond your limits by skipping exercise, having an extra cup of coffee to stay awake, and substituting nutritious meals (ホ) fast food. It is only a matter of time before such trade-offs come back to bite you. (い) in the long run. Set healthy boundaries with yourself. You are human and need to honor your limits.

注 resume 履歴書

hot fudge sundae シロップや果物のソースなどがかったアイスクリーム
の上に温かいチョコレートソースをかけたデザート

- 1 空欄(A)—(E)に入れるのに最も適した語を次の中から選び、それぞれ解答欄に書きなさい。ただし、各語は1回のみ使用できるものとする。

apart away better instead only

- 2 空欄(イ)—(ホ)に入れるのに最も適した語を次の中から選び、それぞれ解答欄に書きなさい。ただし、各語は1回のみ使用できるものとする。

by in like on with

- 3 空欄(あ)に入れるものとして、以下の語を最も適切な順に並べ替えたとき、5番目と9番目に来る語を解答欄に書きなさい(並べ替えたとき最初に来る語の最初の文字も小文字にしてある)。

a are ask considering goal
of particular pursuit the you
yourself why

- 4 空欄(い)に入れるものとして、以下の語を最も適切な順に並べ替えたとき、5番目と9番目に来る語を解答欄に書きなさい(並べ替えたとき最初に来る語の最初の文字も小文字にしてある)。

and care effective efficient less
make not of taking will
you yourself

III Choose one of the questions below and answer it in English. Your answer should be 100 to 140 words in length. Indicate the number of the question you have chosen. *Correctly* indicate the number of words you have written at the end of the composition.

- 1 When you read a book or watch a film, do you focus more on the story, the characters, or the atmosphere?
- 2 When speaking a foreign language, which aspect should people pay most attention to: correctness, creativity, or speed?
- 3 Which do you think is the most valuable quality for a leader: intelligence, ambition, or honesty?

Ⅳ 音声を聞き、その指示に従って、AおよびBの各問いに答えなさい。

A 下の1～6の空欄に入れるのに最も適切なものを選択肢イ～ニからそれぞれ一つ選び、その記号を解答用紙に書きなさい。

1 German small talk style is described as “coconutty” because _____.

- イ Germans prefer private conversations at work
- ロ Germans seem hard on the outside
- ハ it is similar to the American “peach” style
- ニ it mixes professional and personal topics

2 In “coconut” culture, _____.

- イ professionals take time for small talk
- ロ questions about private life are valued
- ハ small talk is not as important as work
- ニ work is not as serious as private matters

3 When meeting a “peach-like” American, _____.

- イ personal small talk indicates a close friendship
- ロ personal small talk shows interest and enthusiasm
- ハ you do not share personal information at first
- ニ you do not smile or ask many personal questions

4 The differences between “coconut” culture and “peach” culture are more likely to _____.

- イ create conflicts between people
- ロ encourage stereotypes
- ハ help people come closer
- ニ make people speak too much

5 In real life, “coconuts” and “peaches” _____.

- イ are fairly simple to understand
- ロ are not stereotypes of people
- ハ can be compared to a cultural bridge
- ニ include many different kinds of people

6 The speaker thinks the “coconut and peach” analogy _____.

- イ makes others adapt to our speech style
- ロ makes others decide which culture is better
- ハ will help us communicate with different people
- ニ will help us know in which business we belong

B 下の1～6の空欄に入れるのに最も適切なものを選択肢イ～ニからそれぞれ一つ選び、その記号を解答用紙に書きなさい。

- 1 Purple tomatoes are _____.
イ commonly found on the shelves of supermarkets
□ created by mixing dark-colored vegetables
ハ designed to not produce “anthocyanins”
ニ not found in nature
- 2 The substance that turns tomatoes purple might _____.
イ add a berry-like flavor
□ cause cancer in mice
ハ help humans live a longer life
ニ keep mice away from tomatoes
- 3 The fact that purple tomatoes have longer shelf life would _____.
イ help consumers control the amount of food thrown away
□ help our bodies stay young for longer
ハ make more people eat tomatoes cooked rather than fresh
ニ speed up the process of approval by the authorities
- 4 According to the speaker, one important difference between seedless fruits and purple tomatoes is _____.
イ the speed and the accuracy of the process used to develop them
□ where the products were first grown
ハ whether the technology made the product easier to eat
ニ whether their mass production happened naturally

5 Purple tomatoes will become available when _____.

- イ technology becomes more advanced
- the authorities decide they are safe
- 入 the developers apply for approval
- ニ the developers set new standards

6 The speaker's conclusion suggests that genetically engineered farm products _____.

- イ should be approved by the authorities
- should be taken into people's diet
- 入 will become common among consumers
- ニ will give suppliers more freedom

